

Mathematical Foundations of Political Methodology

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Statistical Inference

- Model based inference:
- **Assume**: data generated via distributional process
- Defines a likelihood function: parameter values $\rightarrow$ likelihood of parameters, given data
- Derive estimators that identify values that **maximize** likelihood

General Likelihood Theory $\rightarrow$ Example $\rightarrow$ General Theory $\rightarrow$ Example

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We're going to be interested in making an inference about θ_0 using the observed data.

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- $L(\theta|\mathbf{y}) : \underbrace{\Theta}_{\text{parameter space}} \rightarrow \mathbb{R}$
- Idea: values of $\theta \in \Theta$ will be **more likely** if they make the observed data a higher probability

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We also are only able to infer most likely value **given modeling assumptions**

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- $y_i = 1$ or $y_i = 0$

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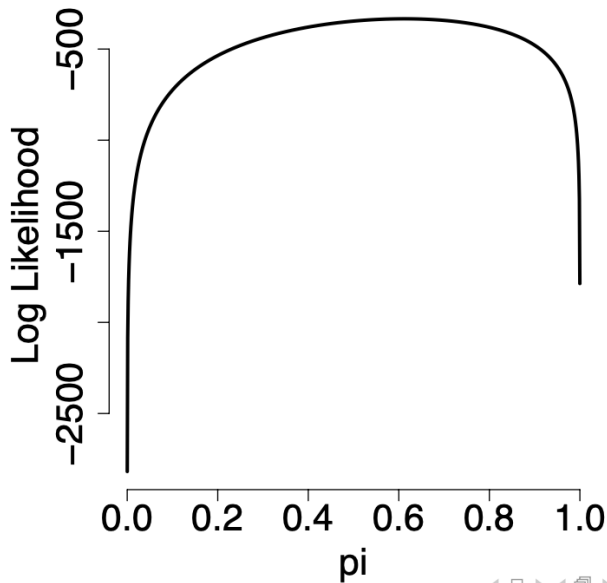
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For a fixed set of observations, what does this look like?

$$\pi = 0.6$$



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$$\pi^* = \bar{y}$$

The **Fisher Information** measures the information that $\mathbf{y}$ conveys about the parameter θ . Define it using the two equivalent definitions:

Definition

The **Fisher Information** for a log-likelihood $l(\theta|\mathbf{Y})$ is

$$\begin{aligned} I(\theta) &= -E \left[\left(\frac{\partial l(\theta|\mathbf{Y})}{\partial \theta} \right)^2 \middle| \theta \right] \\ &= -E \left[\left(\frac{\partial^2 l(\theta|\mathbf{Y})}{\partial \theta \partial \theta} \right) \middle| \theta \right] \end{aligned}$$

The **observed Fisher information** for a sample of n observations is given by

$$I_n(\theta) = -\frac{\partial^2}{\partial \theta^2} l(\theta|\mathbf{y})$$

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Uncertainty About Mode

Inverting the information provides the asymptotic variance for the maximum likelihood estimator (under some regulatory conditions we will discuss later)

$$\text{Variance}(\theta^*) = \frac{1}{I_n(\theta^*)}$$

$$\text{Standard Error}(\theta^*) = \sqrt{\frac{1}{I_n(\theta^*)}}$$

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- **Curvature** determines sampling distribution of maximum likelihood estimator

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- 1) The MLE gets as “close” as possible to the true answer

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Proposition

MLE is consistent: Assume $y_1, y_2, \dots, y_n$ are simple random samples from $p(y|\theta_0)$. Define θ_n^* as the mle estimator with sample size n . Then, as $n \rightarrow \infty$, $\theta_n^* \rightarrow \theta_0$ (in probability)

Definition

$\hat{\theta}_n$, with cdf $F_n(x)$, converges in distribution to random variable Y with cdf $F(x)$ if

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For all $x \in \mathbb{R}$ where $F(x)$ is continuous.

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- MLE central limit theorem
- As we have more observations, the MLE converges, **in distribution** to a normal distribution

Returning to Our Examples

Summary: θ^* is an MLE and θ_0 is the true value and we know the right distributional family

1) As $n \rightarrow \infty$, $\theta_n^* \rightarrow \theta_0$

2) As $n \rightarrow \infty$, $p(\theta^*) \rightarrow \text{Normal}(\theta_0, \frac{1}{I(\theta_0)})$

where $I(\theta_0) = -\frac{\partial^2}{\partial \pi^2} l(\theta_0|y)$ or curvature of log-likelihood at true value of θ_0